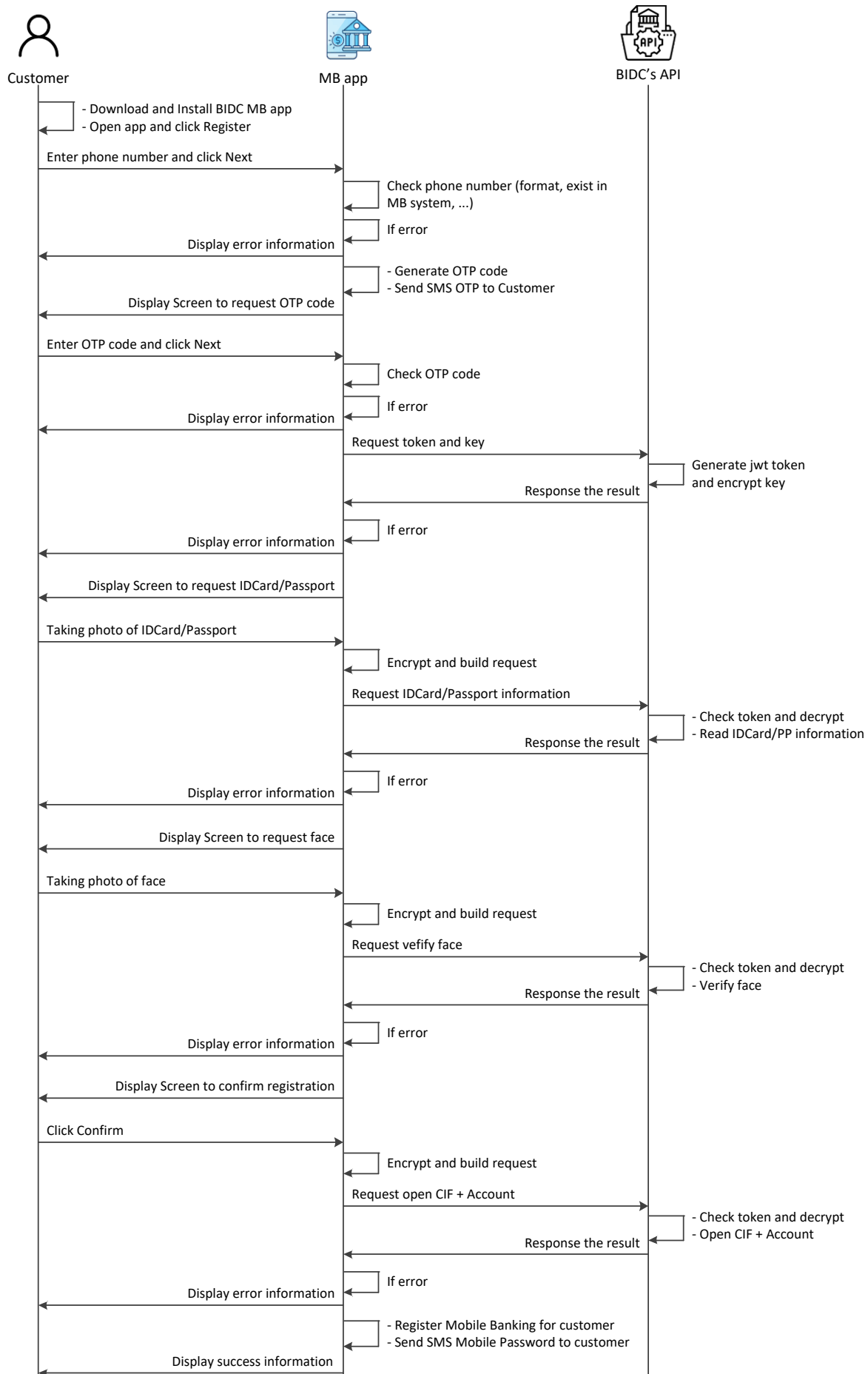


SPECIFICATION FOR EKYC SERVICE ON MOBILE BANKING BIDC

1. Flowchart



2. Message specification

2.1. Initiate a session

- Generate jwt token and key to manage session and security.
- Mobile Banking call to BIDC API through **DBS System**
- **Request/Response:**

URL	http://10.195.101.21:8089/BIDC/DBS_GW/doProcess		
METHOD	Post		
	Param	R/O	Description
REQUEST	Header		
	Content-Type	R	Application/json
	Authorization	R	Basic auth bidc_dbs/A49ljC8rgMaA8GS4CKv35w= =
	Body		
	{ "messageId": "67", //mandatory "channel": "EKYC", //mandatory "deviceModel": "sdk_gphone64_arm64", "deviceId": "UE1A.230829.036", "platform": "Android", "userId": "0885444292", //mandatory "jwtExpireTime": "5" // minutes - mandatory }	R	
RESPONSE	{ "errorCode": "0", "errorDetail": "Success", "tokenJwt": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJJEZXXZpY2VNb2"		

	RlbCI6InNka19ncGhvbU2NF9h cm02NCIsInN1YiI6ImJpZGMuY2 9tLmtoIiwiUGxhdGZvc0iOiJp T1MiLCJEZXZpY2VJZCI6IlVFMU EuMjMwODI5LjAzNiIsIkFwcGxp Y2F0aW9uSWQiOiJFS1lDIn0.nu ll", "key": "lJDvXbEmBHgEsv+U7APyGR8hr 86gDjSSPfAg+auITOM=" }		
STATUS CODE	HTTP STATUS 200: ErrorCode: "0": "SUCCESS"		

2.2. OCR IDCard (support only Khmer ID)

- Read IDCard Information through image.
- Mobile Banking call to BIDC API via **BIDC eKYC API** on internet.
- **Request/Response:**

URL	https://api-ekyc-uat.bidc.com.kh/api/v1/ekyc		
METHOD	Post		
	Param	R/O	Description
REQUEST	Header		
	Content-Type	R	Application/json
	Authorization	R	Bearer Token
	Body		
	<pre>{ "msgId": "01", "requestId": "", //generate at frontside "encryptedString": "", "nonceString": "" }</pre>	R	encryptedString : "{\"imageFront\": \"base64\", \"imageBack\": \"base64\", \"userId\": \"0885444293\"}"

RESPONSE	<pre>{ "errorCode": "00", "errorDetail": "SUCCESS ID READER", "encryptedString": "" }</pre>	<pre>encryptedString:"{ "transId": "17664479744315", "pathFront": "", "pathBack": "", "ekycRef": "EKYC20251228004", "idCardInfo": { "idNumber": "", "fullName": "", "lastName": "", "firstName": "", "isFraud": "", "enDob": "", "enIssuedDate": "", "enExpiredDate": "", "enAddress": "", "enProvince": "", "khDob": "", "khIssuedDate": "", "khExpiredDate": "", "khAddress": "", "khProvince": "", "provinceCode": "CHA", "pathImage": "", "enSex": "Male", "khSex": "" } }"</pre>
STATUS CODE	HTTP STATUS 200: ErrorCode "00": "SUCCESS" "01": "BUSINESS VALIDATION FAILED" "02": "BIDC EXCEPTION" "03": "LEGAL ID already existed" HTTP STATUS 401: Token invalid	

2.3. ID Verify face

- Verify face to check if match with IDCard.
- Mobile Banking call to BIDC API via **BIDC eKYC API** on internet.
- Request / Reponse:

URL	https://api-ekyc-uat.bidc.com.kh/api/v1/ekyc		
METHOD	Post		
	Param	R/O	Description

REQUEST	Header		
	Content-Type	R	Application/json
	Authorization	R	Bearer Token
	Body		
	<pre>{ "msgId": "02", "requestId": "",//generate at frontside "encryptedString":"", "nonceString": "" }</pre>	R	<pre>encryptedString : "{ "image": "", "transId": "17669596925068", "userId": "0885444293", "ekycRef": "EKYC20251228004",//Get from the API OCR Response "pathFront": ""// get from idCardInfo.pathImage }"</pre>
RESPONSE	<pre>{ "errorCode": "00", "errorDetail": "SUCCESS ID READER", "encryptedString": "" }</pre>		<pre>encryptedString:"{ "face": { "match": true, "fraud": false }, "pathFile": "" }" //match = true: image matched with document //match = false: image mismatched with document // fraud: false : image is normall // fraud: true: image is abnormal</pre>
STATUS CODE	HTTP STATUS 200: ErrorCode: "00": "SUCCESS" "01": "VALIDATION FAILED" "02": "BIDC EXCEPTION" HTTP STATUS 401: Token invalid		

2.4. Open CIF and Account

- Open CIF and Account (both USD and KHR).
- Mobile Banking call to BIDC API via **BIDC eKYC API** on internet.
- Request / Response:

URL	https://api-ekyc-uat.bidc.com.kh/api/v1/ekyc		
METHOD	Post		
	Param	R/O	Description
REQUEST	Header		
	Content-Type	R	Application/json
	Authorization	R	Bearer Token
	Body		
	<pre>{ "msgId": "03", "requestId": "",//generate at frontside "encryptedString":""," "nonceString": "" // }</pre>	R	<pre>encryptedString : "{ "userId": "0885444292", "ekycRef": "EKYC20251228004", "title": "MR", "givenName": "BALITON", "familyName": "ROWENA", "shortName": "BALITON ROWENA PACLIBON", "fullName": "BALITON ROWENA PACLIBON", "birhtDate": "19771115", "address": "Takeo", "country": "Cambodia", "province": "03", "district": "0305", "commune": "030502", "village": "03050205", "nationality": "KH", "residence": "KH", "legalDocName": "ID", "legalDocId": "101262302", "legalHolderName": null, "legalIssDate": "20240509", "legalExpDate": "20340508", "gender": "MALE", "phoneNumber": "0112233", "email": null, "occupation": "5", "coCode": "KH0010001" }"</pre>
RESPONSE	<pre>{ "errorCode": "00", "errorDetail": "SUCCESS ID READER", "encryptedString": ""</pre>		<pre>encryptedString:"{ "cifNo": "156939", "fullName": "ROWENA BALITON", "accountUSD": "703700470777", "accountKHR": "703000453693"</pre>

	}		}"
STATUS CODE	HTTP STATUS 200: ErrorCode: "00": "SUCCESS" "01": "VALIDATION FAILED" "02": "BIDC EXCEPTION" HTTP STATUS 401: Token invalid		

2.5. Download File

- Download the image file for mobile banking web backend.
- Mobile Banking backend call to ekyc_file_gw for download file.
- Request / Response:

URL	http://10.195.6.52:8181/download		
METHOD	Post		
	Param	R/O	Description
REQUEST	Header		
	Content-Type	R	Application/json
	Authorization	R	Basic auth bidc-mb / BidcBoneMbEkyc@1231
	Body		
	{ "Type": "ID", "FilePath": "" }	R	-Type: ID -File Path:
RESPONSE	Binary file or { "errorCode": "1", "errorDetail": "FILE NOT FOUND" }		
STATUS CODE	HTTP STATUS 200: "SUCCESS" 401: Token invalid HTTP STATUS 200 ErrorCode: "1": "FILE NOT FOUND"		

3. Encryption and Security

- Reduce the image size to around **1024–1280 px** to decrease file size while keeping OCR quality high
- Use JPEG compression to keep the file lightweight (300–600 KB) without losing important details
- Convert the optimized image into Base64 for safe and consistent transmission to the API
- Encrypt the Base64 data with AES-256 before sending.
- Request sent to **BIDC eKYC API** must include JWT-based Authorization Token
- JWT token and encrypt key (for AES-256) were generated when Initiate a session.